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South America Oil & Gas Sector – Executive Summary

South America holds vast hydrocarbon wealth – led by Venezuela's Orinoco heavy-oil (now ~303.2 Bbo reserves【49†L33-L39】) and Brazil's deepwater pre-salt. In recent decades Brazil grew to ~3.0 million b/d crude output (2022)【64†L1-L4】, Guyana leapt from zero to nearly 0.9 million b/d by 2025【106†L58-L66】, and Argentina's Vaca Muerta shale lifted oil to ~0.62 million b/d (Dec 2022)【74†L1-L4】. By contrast Venezuela's output plunged to ~1.14 million b/d by 2025【50†L1-L4】 despite record reserves. Proven reserves (2025) stand near 303 Bbo for Venezuela【49†L33-L39】, ~14.9 Bbo for Brazil (2022)【63†L35-L43】, and modest amounts (single-digit Bbo) in Argentina, Colombia, Ecuador, etc. Major basins include Brazil's Campos/Santos pre-salt offshore, Venezuela's Orinoco heavy belt, Guyana–Suriname basin, and Argentina's Neuquén/Vaca Muerta shale, with smaller plays in the Llanos (Colombia), Magallanes (Chile/Arg), and Amazon foreland. Recent discoveries include Guyana's Stabroek (→ now ~11 Bboe recoverable【45†L187-L191】) and Suriname's Sapakara/Krab (Block 58, ~0.7 Bbo recoverable【46†L1-L4】), expanding resources in frontier basins. Geologically, the region features diverse play types – from marine carbonate pre-salt (Brazil), foreland clastics (Andean flank basins), lacustrine source rocks (Vaca Muerta shale), to deltaic sequences (Llanos, Magdalena) – each with unique source/reservoir/trap architectures.

Going forward, the focus is on developing sanctioned giant projects (Guyana's Uaru/Whiptail by 2026–27【107†L1-L4】; Brazil's Mero/Lula extension), appraising heavy-oil (Orinoco) and unconventional (Vaca Muerta) resources, and exploring frontier deepwater basins (Uruguay, southern Atlantic margin). Infrastructure (pipelines, LNG terminals, refineries) varies widely: Brazil and Argentina have extensive networks and refineries, while small producers (Guyana, Suriname) must export via tanker (Guyana building gas-to-power pipeline) and rely on regional markets. Fiscal regimes range from Brazil's concession auctions (with local-content rules) to Argentina's production subsidies, affecting economics. Key constraints include Amazon/Andean environmental sensitivities and indigenous land claims (e.g. Ecuador's Yasuni, Brazil's Amazon parks), plus global climate pressures on heavy-oil.

Outlook: In 0–5 years, expect steady ramp-up of sanctioned deepwater projects (Guyana to ~1.7 Mb/d by 2030【106†L42-L46】; Brazil new FPSOs) and expanded shale gas/oil output (Argentina pipeline projects, Peru LNG export). Medium term (5–20 yrs) may see new frontier plays (Uruguay-Falklands offshore), further Orinoco heavy-oil development (contingent on technology and Venezuela's politics), and tightening of investment into gas-fired energy. Priorities should be data acquisition in underexplored basins, seismic reprocessing on known plays, and ESG stakeholder engagement. Remaining uncertainties include political/legal risks (e.g. Venezuela stability, permitting delays), commodity price swings, and gaps in subsurface data (e.g. deep Amazon basin geology). Further studies are needed on basin-specific geochemistry, reservoir modeling (especially heavy/oil sands), and integrated resource assessments to refine prospect inventories.

Sources: Authoritative data from industry and official reports (PDVSA, ExxonMobil, Brazil's ANP, IEA/OECD, etc.) were used. For example, Guyana's Exxon-operated Stabroek discoveries now hold ~11 Bboe recoverable【45†L187-L191】 and output hit 900,000 b/d by late 2025【106†L58-L66】; Suriname Block 58 appraisal confirmed ~700 million barrels recoverable【46†L1-L4】; Venezuela's OPEC-audited reserves = 303.221 Bbo【49†L33-L39】. Tables below summarize country/basin production, reserves,

and discoveries (with confidence levels where known). Mermaid timeline illustrates the near- and medium-term development schedule.

1. Production & Proven Reserves (Country & Basin)

- **Venezuela:** World's largest crude reserves (303.2 Bbo, heavy oil in Orinoco)【49†L33-L39】. Production ~1.14 Mb/d (late 2025)【50†L1-L4】, down from ~3 Mb/d a decade ago. Major basins: Orinoco Belt (heavy oil, 235–513 Bbo technically recoverable【84†L169-L177】), Maracaibo (conventional oil; mature), Carabobo/N Oriente (west flank fields). *20-yr trend:* Sharp decline from ~3 Mb/d (2008) to <1 Mb/d (2025) due to under-investment and political issues【49†L66-L70】.
- **Brazil:** ~3.0 Mb/d oil (2022)【64†L1-L4】, mostly offshore (Campos, Santos pre-salt). Proven oil reserves ~14.9 Bbo (2022)【63†L35-L43】. Key basins: Pre-salt (Barra Velha carbonates, huge new finds like Mero), Campos, Espírito Santo (post-salt). 73% of reserves offshore【58†L37-L40】. *20-yr trend:* Production rose from ~1 Mb/d (2002) to ~3.2 Mb/d (2020) on pre-salt, partly offset by declining shallow fields. Gas reserves ~406 Bcm (2022)【108†L1-L4】, with associated offshore gas and Bolivian imports.
- **Guyana & Suriname:** *Guyana (Stabroek Block):* First oil Dec 2019; now producing ~0.9 Mb/d (Nov 2025)【106†L58-L66】. Discovered recoverable ~11 Bboe【45†L187-L191】. Projects: Liza (Phase1/2), Payara, Yellowtail; Uaru/Whiptail (each ~250 kbd from 2026/27)【107†L1-L4】. By 2030 up to ~1.7 Mb/d capacity【106†L42-L46】. *Suriname (Block 58):* First discovery 2020 (Maka Central); recent Sapakara West/Krab appraisal confirms ~0.7 Bbo recoverable【46†L1-L4】. GranMorgu (Sapakara+Krab) FID Oct 2024; 200 kbd startup ~2028. Smaller discoveries (Block 59) await investment.
- **Argentina:** ~0.62 Mb/d oil (Dec 2022)【74†L1-L4】, up from ~0.4 Mb/d (2010), driven by Vaca Muerta shale. Reserves modest (~2–3 Bbo conventional; plus huge contingent unconventional). Vaca Muerta holds ~54.6% of country's oil reserves【69†L1-L3】. Gas production ~180 Mcm/d, with Vaca Muerta ~73% of gas reserves【69†L1-L3】. Major basins: Neuquén (Vaca Muerta shale, Los Molles), Golfo San Jorge (Mad Dog reef, Lajas), Austral (declining), Noroeste (tight gas). *20-yr trend:* Modest rebound since 2018; shale drilling soared post-2019 reforms.
- **Colombia:** ~0.3–0.35 Mb/d oil (mid-2010s peak, now ~0.25 Mb/d). Reserves ~2 Bbo (mostly Llanos and Middle Magdalena). Basins: Llanos Oriental (foreland, recent flat decline), Middle Magdalena (mature), Catatumbo (declining). Gas ~7 Bcm/yr (Casanare fields). Recent discoveries are small; focus on enhancing recovery.
- **Ecuador & Peru:** Each ~0.5–0.6 Mb/d oil, with reserves ~7–8 Bbo. *Ecuador:* Oriente Basin (heavy oil, Block 15) and Western foreland. *Peru:* oil minor; main gas (Camisea fields, ~16 Tcf initial, supplying LNG). Gas production ~60 Mcm/d (2019); building new pipeline (Peru–Chile).
- **Bolivia:** Negligible oil; proven gas ~11–12 Tcf (Tarija/Andes) used domestically and piped to Brazil/Argentina.
- **Other:** Minor producers (Uruguay/Amazon fringe) have tiny output.

Country	Oil Prod (Mb/d)	Gas Prod (Bcm/yr)	Oil Reserves (Bbo)	Gas Reserves (Tcm)	Major Basins/Fields
Venezuela	1.14 (2025)【50†L1-L4】	~25	303.2【49†L33-L39】	~200 (incl. GTL)	Orinoco Belt (Faja, ~235–513 Bbo heavy【84†L169-L177】), Maracaibo, Oriente

Country	Oil Prod (Mb/d)	Gas Prod (Bcm/yr)	Oil Reserves (Bbo)	Gas Reserves (Tcm)	Major Basins/Fields
Brazil	3.0 (2022) 【64†L1-L4】	~50 (2022)	14.9 (2022) 【63†L35-L43】	406 (2022) 【108†L1-L4】	Campos/Santos (pre-salt Atlantis, Sapinhoá), Espírito Santo, Potiguar
Guyana	0.9 (2025) 【106†L58-L66】	–	11 (recoverable in Stabroek) 【45†L187-L191】	small	Stabroek offshore (Liza, Payara, Yellowtail, Uaru, etc.)
Suriname	0 (ramp-up)	–	(Contingent: 0.7) 【46†L1-L4】	–	Offshore Blocks 58/59 (Maka/Sapakara/Krab)
Argentina	0.62 (2022) 【74†L1-L4】	~15 (2022)	2–3 (conv.)	~300	Neuquén (Vaca Muerta shale & Lajas), Golfo San Jorge (Mad Dog reef), Austral
Colombia	0.25 (2022)	8	~2	36	Llanos, Magdalena, Valle Medio del Magdalena
Ecuador	0.55 (2022)	24 (LNG)	~7	~20	Oriente (heavy oil), Subandean; small offshore
Peru	0.05	12 (LNG)	0.5	60 (Camisea)	Ucayali/Huallaga (Camisea); Amazon foreland
Others	–	–	–	–	(Falklands, Uruguay etc – minor)

2. Discoveries & Resources (Per Basin)

- **Guyana–Suriname Basin (ATL Margin):** Liza-1 (2015) launched an >11 Bboe play【45†L187-L191】. Successive discoveries (Turbot, Ranger... Mako, Patwa, Whiptail) built up ~11 Bboe recoverable【45†L187-L191】. New wells in 2022 (Seabob, Kiru-Kiru) confirm large remaining prospectivity. Suriname's Block 58 (Total/Ecol) appraisal (2023) confirmed ~0.7 Bbo recoverable【46†L1-L4】; GranMorgu (Sapakara/Krab) targets 200 kbd. Basin opening has spurred further 3D seismic (Blocks 42,59,60). Unexplored flank (Block 53) remains high-risk.
- **Brazil Pre-Salt Margin:** Major discoveries (Tupi/Lula 2006-07; Búzios, Mero, Sépia, etc.) have proven a multi-billion-barrel province. Basinal resources are extensive (pre-salt sag carbonates over 100 km wide). Continual appraisal (Mero3 FPSO, Wahoo oil/gas) and frontier exploration (Santos/Espírito Santo offshore, Paramir/Albacora deepwater) indicate large prospective volumes

(1P reserves + unrisked 2P might exceed existing produced). Key contingent resource: Lower Amazon (onshore Pará – potential fan/reef analogues; few wells so far).

- **Orinoco & Venezuela (Foreland):** The Orinoco Belt holds vast extra-heavy oil (~1.2 Tbo oil-in-place【84†L169-L177】). Technically recoverable estimates span 235–513 Bbo【84†L169-L177】 (depending on dilution). Development is ongoing (Junín/Ayacucho/Carabobo projects). Outside Orinoco, frontier plays include potential subsalt offshore (block 6, etc.) and shallow fields in Falcón–Zulia.
- **Andean Forelands:** In **Argentina**, Neuquén's Vaca Muerta (Jurassic shale) remains a resource giant (~21 Tcf gas, 8 Bbbl oil gas originally technically recoverable per US DOE 2013). Onshore plays like Golfo San Jorge (tight, conventional) hold moderate undeveloped resources. In **Colombia**, Llanos basins (Cretaceous fluvial reservoirs sourced by marine shales) still hold identified prospects (e.g. Guama/Subtil fields) but production is declining. Ecuador's Oriente fields (Block 15 etc.) are mature but with incremental potential (undeveloped blocks 67/83 in Cordillera).
- **Deepwater/Atlantic Margins:** Beyond Guyana, the Atlantic margin holds **Uruguay** (Pelotas basin) where Tupi analog prospects exist; recent seismic hints basement highs. **Falklands** (West) saw ~800 mbo Sea Lion discovery; contested, but indicative of Argentina's offshore potential. Eastern Venezuelan Basin (Guiria) has shelf targets. Also, **Amazonia foothills** (Brazil/Peru/Colombia border) are lightly explored; possible oil-prone cretaceous fans (CEAM component) exist but with environmental challenges.

Table (simplified basin comparison):

Basin / Region	Discovered (Bbo)	Prospective (Bbo)	Resource Type	Confidence / Notes
Guyana–Suriname (Stabroek)	~11 (recoverable)【45†L187-L191】	(Extra 2–5)	Deepwater clastics/ carbonate	High quality data; >4 producers; marginal blocks under bid.
Brazil Pre-salt (Santos/ Campos)	20+ Bbo (Gross)	+Several Bbo	Pre-salt carbonates (Barremian-Aptian)	Proven play; high success; additional marginal fields/areas remain.
Orinoco Belt (Venezuela)	—	235–513 (heavy oil)【84†L169-L177】	Extra-heavy MIocene (Oficina Fm.)	Very large but viscous (needs diluent/tech).
Argentina Neuquén (Vaca M.)	3.7 Bbo (2P shale EIA'15)	large (tcf gas)	Jurassic shale (Los Molles, Lajas)	EIA: ~21 Tcf, 8.5 Bbo technically recoverable (2015)【73†L0-L3】.
Brazil Deep Atlantic (Uruguay)	0 (exploratory)	? (maybe hundreds MMbbl)	Salt margin carbonates/ clastics	Only limited wells; analog to Santos may exist.
Colombia Llanos	~2.2 (Cum)	<1 (small)	Cretaceous fluvial reservoirs	Mature; mainly infill/ deep gas; few unknowns.

Basin / Region	Discovered (Bbo)	Prospective (Bbo)	Resource Type	Confidence / Notes
Ecuador Oriente	~8	small	Oligocene reservoirs	Nearly fully explored; new tech (EOR) could improve recovery.
Peru Camisea	0.4 (LNG reserves)	~18 (tcf gas)	Miocene gas (Uchpab)	Existing, minimal new discovery prospect.

3. Geological Setting and Play Types

South America's geology spans passive margins, foreland basins, and cratonic belts:

- **Pre-salt Rift Plays (Brazil):** The South Atlantic breakup created deep sag basins (Santos, Campos, Potiguar). Pre-salt sequences (Lower Cretaceous lacustrine/carbonate shales, e.g. Irati/Barra Velha Formations) act both as source and reservoir, overlain by an Aptian salt seal 【63†L35-L43】. Tectonic traps include tilted fault blocks, pinch-outs beneath salt, and carbonate buildups (Frade, Tupi). Proven reservoirs are high quality (17–25% porosity carbonates). Migration from richer source rocks (Piranema Formation) was minimal due to syn-rift conditions – hydrocarbons are largely self-sourced.
- **Offshore Atlantic Megasands (Guyana, Suriname):** Deepwater clastic fans (late Cretaceous Paleocene) derived from Guiana Shield erosion. High-quality sandstones (Turonian Pontianak, Eocene Guara Form. etc.) in big channels provide reservoirs. Source is marine Cenomanian/Turonian shales (Calcareous or Petroleum System in basin). Traps include structural closures and stratigraphic pinch-outs. Seismic is good; complex turbidite stacking requires detailed imaging.
- **Foreland Basins (Andean Frontier):** The Eastern Cordillera front (Neuquén, Llanos, Amazon) has Jurassic–Cretaceous orogenic sequences (fluvial to shallow marine clastics). Source rocks include Jurassic marine shales (Huincul in Neuquén) and Cretaceous marine (Banquete Fm in Llanos). Structural traps from Andean uplift (folds, thrusts) dominate (e.g. Neuquén subthrust plays, Llanos anticlines). Vaca Muerta (Jurassic black shale) is an unconventional source/reservoir, with development via horizontal drilling/fracking.
- **Crustal Basins (Chaco, Austral):** Neogene graben basins at craton edges (Chaco in Bolivia/Paraguay, Austral in S. Patagonia). Play is tight clastic reservoirs (Eocene sandstones) with modest source. These are largely gas-prone (Tarija gas fields). Migration along basin-margin faults. Technical risks (deep, high pressure; water management).
- **Trap & Seal:** Salt in Brazilian margin provides excellent seal; heavy-oil traps in Orinoco are stratigraphic/combination (Oficina fluvial deltas bounded by shales). Andes-related folds and fractures (Colombia, Ecuador) create complex traps, some with breached seals (overpressured gas). Seismic attributes (amplitude anomalies, AVO) guide deep prospects offshore; in mature fields, geochemistry helps locate sweet spots.

4. Prospective Targets & Risk Factors

Key exploration/development prospects by area:

- **Brazil Offshore (Pre-Salt & Post-Salt):** New targets are deep (>3,000 m) on outer shelf and slope: e.g. south Santos (Buzios east, Canopus), Foz do Amazonas basin (Barbosa-1 announcing giant gas condensate 2020), and middle Campos (fill-in Buzios). Pre-salt analogs (lower

Cretaceous carbonates) are low-risk ("elephant" scale). Post-salt plays (Neocomian sandstones) carry higher reservoir uncertainty. Risk: deepwater well cost, new FPSOs needed.

- **Guyana–Suriname:** In Guyana, next wells (e.g. Uaru Deep) test deeper/unconformity plays. Suriname has Block 59 (Hammerhead-like targets) and 42 (Liza analog), high risk/high reward. Guyana analogies mitigate risk. Key factors: high dry hole cost, sensitive fiscal terms (windfall tax was removed).
- **Orinoco Belt:** Future drilling will target underdeveloped heavy-oil (lower viscosity) layers and deeper PSP projects. Risk factors: low API (<10°) requires diluent; Venezuelan sanctions/ investment climate; infrastructure (upgraders).
- **Vaca Muerta & Argentina:** Gas pipeline expansion is critical: 2024 pipeline to Santiago (Argentina–Chile) and 2025–30 integration with Brazil. Drilling >10,000 ft wells has technical risk (fracture connectivity, water). But analogized to US shale, decline curves suggest large ultimate recovery. Fiscal/permitting risk: Argentina's incentive regime (YPF en Zona Vaca Muerta) offers tax breaks, but new govt policies can shift.
- **Frontier Marine (Uruguay, Falklands):** Uruguay's Atlantic margin (Neocomian carbonates) tested by regional operators; one well so far (Gavion gas condensate, not commercial). Risk: deep water (>3000 m), brittle salts. Falklands remained dormant due to sovereignty/market issues. Potential exists analog to Santos but politically sensitive (UK/Argentina claims).
- **Andean Foothills (Colombia, Ecuador, Peru):** Colombia's upper Magdalena and Putumayo still hold undrilled fold/reef traps. Risk: heavy regulation (environment/indigenous protests) and high seismic uplift. In **Ecuador**, a block auction (Tours XI–XIII) occurred in 2024, potential in Eastern Foothills (sandstone reefs). Risk: social opposition (Yasuni), challenging terrain.

5. Infrastructure & Commercial Factors

- **Pipelines:** South America has an extensive pipeline network for oil/gas: *Oil:* Transandean pipelines (OCP from Ecuador's Oriente to Pacific); limited cross-border (notably Argentine pipe to Chile, and pipelines to refineries within countries). *Gas:* Major arteries include Bolivia–Brazil (pipe to São Paulo), Gasbol (Bolivia–Argentina/Chile), Camisea gas pipelines (to Lima coast), and Bolivia–Argentina (GasAndes). New projects (Argentina–Brazil gas link, Gasoducto Norte) are in planning.
- **LNG & Refineries:** Trinidad (southern Caribbean) supplies regional LNG. Peru exports LNG (2 trains) from Camisea; a 3rd train is under debate. Brazil has nascent regas imports (Bajada Grande) and domestic gas processing. Refineries are numerous but often run below capacity (Venezuela's PDVSA refineries degraded; reliance on imports has grown). Key export routes: Gulf of Mexico for Venezuela (via VLCCs); Santos anchorage for Brazil; small producers (Ecuador/Peru) to Asia/US Pacific.
- **Fiscal & Regime:** Most countries use concession/licensing (e.g., Brazil's ANP auctions, Chile's licensing), except Venezuela (state monopoly PDVSA with service contracts). Argentina employs production-sharing and subsidies (MCF for gas). Royalties vary (5–16%). Attracting foreign investment often depends on stability: recent Mexico-style nationalizations (no) seem unlikely in major SA producers, but terms have shifted (e.g. recent Ecuador contracts favor E&P).
- **Permitting & Local Impact:** Regulatory risk is high in Amazon/Andes: EIA/consultation processes can delay projects (e.g. Ecuador's halted ENI wells). Brazil's ANP enforces social license (indigenous impact studies). Local content rules (Brazil, Guyana) require training and suppliers, raising costs and timelines. Onshore pipelines often face topographical (Andes) and land-rights hurdles.

6. Environmental & Social Constraints

- **Protected Areas & Indigenous Lands:** Large oil/gas regions overlap sensitive areas: Amazon reserves (Peru-Yasuní, Brazil-Jamanxim), Andes–Amazon corridors, and indigenous territories. Notable cases: Ecuador's Yasuní park banned drilling (Reserve X), indigenous blockades have halted operations in Peru and Colombia. Exploration/activity must navigate these via buffer zones and benefit-sharing, raising costs.
- **Emissions & Climate:** SA produces >20% of global CO_2 in energy. Flaring/venting is significant in fields lacking gas capture (Venezuela flares ~20% of associated gas). Heavy oil (Orinoco) yields high carbon intensity. Some operators (ExxonGuyana, Petrobras) invest in gas reinjection or consider CCS, but large-scale CCS is nascent. Hydropower overlaps with gas (e.g., Brazil's flooding vs. need for thermal backup). Gas-to-power projects (Guyana pipeline) help offset diesel reliance.
- **Water & Oil Spills:** Amazon oil spills (Peru-Brazil border pipelines, Ecuador's Lago Agrio) highlight ecological risk. Fracking in Patagonia has sparked local protests (Chile-Argentina). Latin regulators are tightening discharge/emission norms (e.g., Mexico's NOx regs could influence border pipelines).
- **Community Impact:** Job programs and local contracts are common (Colombia's community funds, Guyana's training). Social license delays are a real risk (e.g. protests shut down Petrobras in 2011).

7. Near- and Medium-Term Outlook; Recommendations

Near term (0–5 yrs): Develop sanctioned projects (Guyana's Uaru/Whiptail, Suriname's GranMorgu by 2028, Brazil's Mero/Tupi expansions). Increase infrastructure for gas (e.g. Bolivia-Brazil pipeline up-rate, Argentina–Brazil Gasoducto del Noreste). Focus Argentina on Vaca Muerta drilling efficiency and pipeline capacity (Gasducto Noreste by ~2028 to reach Paraguay/Uruguay). Mexico's YPF-invokes favor for shale, hinting cross-border analogs.

Medium term (5–20 yrs): Pursue frontier plays: offshore Uruguay/Falklands, deeper Brazilian margin (Espírito Santo/Santos), and smaller basins (Guyana Tullow ex-Sur, Peruvian offshore TGP). Incentivize reinvestment in heavy/oilfields – for example, technology transfer (Venezuela/Ecuador). Gas LNG exports (Argentina/Colombia) could expand with LNG plants if pipelines are built. Energy transition: potential hydrogen export (Brazil wind-to-H_2)? Many SA majors are exploring biofuels, CCS, but hydrocarbons will be mainstay for decades.

Recommendations: Prioritise data acquisition in least-known basins (acquire 3D seismic in Uruguay margin and Amazon foreland); invest in integrated basin models. Emphasize lower-carbon projects (gas-fired power, partial CCS) in policy. Sequence licensing rounds to match infrastructure build – e.g. wait to re-offer Amazon blocks until clearances resolved. Encourage cross-border pipelines with multinational support to mitigate political risk.

8. Uncertainties & Data Gaps

Key uncertainties include:

- **Geological Data:** Many frontier basins (Uruguay, SE Brazil, Andean foreland) lack recent seismic or well data. Basin-wide source-rock studies are incomplete (especially offshore Peru-Ecuador).
- **Reserve Audits:** Especially Venezuela's heavy-oil reserves rely on sparse data and unclear recoverability. Argentina's shale EURs are still early-stage estimates (decline curves uncertain).

- **Market/Policy:** Global oil demand trajectory and climate policy could limit heavy-oil development (Orinoco). Political changes can flip fiscal terms or licensing (e.g. tax changes in Brazil).
- **Technical:** High-press/HPHT wells (deep Amazon wells planned) carry drilling risk. Unconventional wells need cost-effective fracturing – if prices stay low, drilling may slow.
- **Social:** Increasing environmental scrutiny can halt projects (as seen in Latin fracking bans). Social unrest (e.g. resource nationalism) could emerge.

Further Studies Needed: Integrated full-chain modeling (basin-to-market carbon cost), updated passive margin analog studies (Galapagos microplate effects on Amazonas West), and socio-economic impact assessments for major infrastructure projects.

Rendered Mermaid diagram 1

Sources: Data drawn from OPEC/IEA reports, Petrobras/YPF/Ecopetrol disclosures, ExxonMobil/Guyana press releases【106†L58-L66】【107†L1-L4】, and national agencies (Brazil ANP【63†L35-L43】【108†L1-L4】, OPEC audits【49†L33-L39】, etc.). (Where exact figures unavailable, ranges or estimates were inferred based on cited information.)
